

Regression I

Tasks (Lab 12):

1. Generate artificial data using the following model:

$$y = g(x) + \varepsilon,$$

where

$$g(x) = 4.26(e^{-x} - 4e^{-2x} + 3e^{-3x}),$$

and $\varepsilon \sim N(0, 0.1^2)$. Assume that x_i are generated independently from the uniform distribution on $[0, 4]$.

Generate both a training set and an independent test set.

2. Implement from scratch the Nadaraya-Watson kernel regression estimator:

$$\hat{g}_h(x) = \frac{\sum_{i=1}^n K\left(\frac{x-x_i}{h}\right) y_i}{\sum_{i=1}^n K\left(\frac{x-x_i}{h}\right)}.$$

Use at least the Gaussian kernel. You may also compare it with another kernel.

3. Compare the performance of two regression methods:
 - Nadaraya-Watson kernel regression,
 - smoothing splines regression.
4. For the Nadaraya-Watson estimator, compare several values of the bandwidth parameter h . For smoothing splines, compare several values of the smoothing parameter. Explain what happens when the fitted curve is too rough or too smooth. Relate your observations to the bias-variance tradeoff.
5. Visualize the results. Prepare a scatter plot of the training data and draw the curves corresponding to:
 - the true regression function $g(x)$,
 - the fitted Nadaraya-Watson estimators,
 - the fitted smoothing splines.

6. For every considered method and value of the smoothing parameter, compute two types of errors:

- function estimation error:

$$MSE_g = \frac{1}{n} \sum_{i=1}^n [g(x_i) - \hat{g}(x_i)]^2,$$

- prediction error:

$$MSE_y = \frac{1}{n} \sum_{i=1}^n [y_i - \hat{g}(x_i)]^2.$$

Compute both errors on the training set and on an independent test set.

Explain the difference between these two quantities. Which one is available only because this is a simulation experiment? Which one would be available for a real regression dataset?

7. Explore how the test error depends on the training sample size n . Consider several values of n , repeat the experiment several times for each value, and generate a plot showing the average test error as a function of n .

Does increasing n help equally for all smoothing parameter values? Explain why, for the Nadaraya-Watson estimator, the bandwidth h should usually decrease as n increases. What can go wrong if h is kept too large? What can go wrong if h decreases too quickly?

8. Which method is more local and which one is more global? How does a single new observation affect the fitted curve in Nadaraya-Watson regression and in smoothing spline regression?
9. Based on your experiments, explain why choosing the smoothing parameter is a model selection problem. Would you choose the parameter using training error, test error, or cross-validation? Justify your answer.